

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Renee Vincent ASR

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Given:

- File size = **150 MB** = $150 \times 10^6 = 150,000,000$ bytes
- Segment size = **1500 bytes**
- Data Link Layer overhead = **40 bytes/frame**
- Bandwidth = **50 Mbps** = 50×10^6 bits/s

a) Number of Segments

$$\text{Number of Segments} = \frac{150,000,000}{1500} = 100,000$$

Number of Segments = 100,000

Since each segment is encapsulated into one frame,

Number of Frames = 100,000

b) Total Data Link Layer Overhead

Overhead per frame = **40 bytes**

$$\text{Total Overhead} = 100,000 \times 40 = 4,000,000 \text{ bytes}$$

Total Overhead = 4,000,000 bytes (4 MB)

c) Total Transmission Time

Total data transmitted:

$$150,000,000 + 4,000,000 = 154,000,000 \text{ bytes}$$

Convert to bits:

$$154,000,000 \times 8 = 1,232,000,000 \text{ bits}$$

Transmission time:

$$\frac{1,232,000,000}{50 \times 10^6} = 24.64 \text{ seconds}$$

Estimated Transmission Time = 24.64 seconds

d) Reliability Provided by the Layers

Transport Layer

- Divides the file into segments.
- Provides error recovery using acknowledgments and retransmissions.
- Ensures reliable and ordered delivery through flow control and sequencing.

Data Link Layer

- Encapsulates each segment into a frame.
- Detects transmission errors using CRC/checksum.
- Controls access to the physical medium and retransmits corrupted frames.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = **6 frames**
- Data per frame = **1024 bytes**
- Total frames to transmit = **48 frames**

a) Number of Transmission Windows

$$\text{Number of Windows} = \frac{48}{6} = 8$$

Number of Transmission Windows = 8

b) Total Number of Retransmissions

One frame is lost in **every window**.

Number of windows = **8**

$$\text{Total Retransmissions} = 8 \times 1 = 8$$

Total Retransmissions = 8 frames

c) Flow Control and Communication Efficiency

Flow control ensures that the sender does not transmit data faster than the receiver can process it.

- Prevents receiver buffer overflow.
- Reduces packet loss and unnecessary retransmissions.
- Improves network utilization by matching the sender's transmission rate with the receiver's capacity.
- Provides reliable and efficient data transfer using mechanisms such as the **sliding window protocol**.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Given:

- IP packet size = **12,000 bytes**
- MTU = **1500 bytes**
- IP header size = **20 bytes**

a) Maximum Data per Fragment

Each fragment must include a **20-byte IP header**.

$$\text{Data per Fragment} = 1500 - 20 = 1480 \text{ bytes}$$

b) Number of Fragments

$$\text{Number of Fragments} = \left\lceil \frac{12000}{1480} \right\rceil = \lceil 8.108 \rceil = 9$$

Number of Fragments = 9

c) Total Header Overhead

Each fragment has a **20-byte IP header**.

$$\text{Total Header Overhead} = 9 \times 20 = 180 \text{ bytes}$$

Total Header Overhead = 180 bytes

d) How the Network Layer Ensures Successful Delivery

The **Network Layer (IP)** ensures successful delivery of fragmented packets by:

- Dividing large packets into smaller fragments that fit the MTU.
- Assigning each fragment the same **Identification** field.
- Using **Fragment Offset** and **More Fragments (MF)** fields to indicate the position and sequence of fragments.
- Routing each fragment independently to the destination.
- Reassembling all fragments at the destination using the Identification and Fragment Offset fields to reconstruct the original packet.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Given:

- Sliding window size = **6 frames**
- Data per frame = **1024 bytes**
- Total frames transmitted = **48 frames**

a) Number of Transmission Windows

$$\text{Number of Windows} = \frac{48}{6} = 8$$

Number of Transmission Windows = 8

b) Total Number of Retransmissions

One frame is lost in each transmission window.

$$\text{Total Retransmissions} = 8 \times 1 = 8$$

Total Retransmissions = 8 frames

c) Flow Control and Communication Efficiency

Flow control regulates the rate at which the sender transmits data so that the receiver is not overwhelmed.

- Prevents receiver buffer overflow.
- Reduces packet loss and retransmissions.
- Ensures efficient utilization of network bandwidth.
- Improves reliable and smooth data transmission using the sliding window protocol.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Given:

- Video file size = **600 MB**
- Synchronization checkpoint = **every 60 MB**
- Failure occurs after **390 MB**

a) Number of Checkpoints Created Before Failure

Checkpoints are inserted after every **60 MB**.

Checkpoints occur at:

- 60 MB
- 120 MB
- 180 MB
- 240 MB
- 300 MB
- 360 MB

The next checkpoint (420 MB) is not reached before the failure.

Number of Checkpoints Before Failure = 6

b) Data to be Retransmitted After Recovery

The last successful checkpoint is at **360 MB**.

Failure occurs at **390 MB**.

$$390 - 360 = 30 \text{ MB}$$

Data to be Retransmitted = 30 MB

c) Importance of Synchronization in the Session Layer

The **Session Layer** provides synchronization by inserting checkpoints during data transmission.

- Enables communication to resume from the last checkpoint after a failure.
- Avoids retransmitting the entire file.
- Saves transmission time and network bandwidth.
- Improves reliability and efficiency in long data transfers.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Given:

- Original file size = **900 MB**
- Compression = **40%**
- Encryption overhead = **5% of compressed size**

a) Compressed File Size

Compression reduces the file size by **40%**.

$$\text{Compressed Size} = 900 \times (1 - 0.40) = 900 \times 0.60 = 540 \text{ MB}$$

Compressed File Size = 540 MB

b) Encrypted File Size

Encryption adds **5%** overhead.

$$\text{Encryption Overhead} = 540 \times 0.05 = 27 \text{ MB}$$

$$\text{Encrypted File Size} = 540 + 27 = 567 \text{ MB}$$

Encrypted File Size = 567 MB

c) Total Percentage Reduction

Reduction compared to the original file:

$$900 - 567 = 333 \text{ MB}$$

$$\text{Percentage Reduction} = \frac{333}{900} \times 100 = 37\%$$

Total Percentage Reduction = 37%

d) Role of the Presentation Layer

The **Presentation Layer** improves communication by:

- **Compressing data** to reduce file size, saving bandwidth and transmission time.
- **Encrypting data** to ensure confidentiality and protect information from unauthorized access.
- Performing **data format translation** so different systems can exchange data correctly.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Given:

- Total data = **3 GB = 3000 MB**
- Request size = **3 MB**
- Effective throughput = **120 Mbps**

a) Total Number of Requests

$$\text{Number of Requests} = \frac{3000}{3} = 1000$$

Total Number of Requests = 1000

b) Total Transmission Time

Convert data size to bits:

$$3000 \text{ MB} \times 8 = 24,000 \text{ Mb}$$

Transmission time:

$$\text{Time} = \frac{24,000}{120} = 200 \text{ seconds}$$

Estimated Transmission Time = 200 seconds

c) Role of the Application Layer

The **Application Layer** provides reliable services to users by:

- Providing services such as **FTP, HTTP, SMTP, and DNS**.
- Managing file transfer requests and responses.
- Supporting user authentication and access control.
- Detecting application-level errors and ensuring successful file transfer.
- Providing a user-friendly interface for network communication.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Given:

- Application Layer delay = **4 ms**
- Transport Layer delay = **3 ms**
- Network Layer delay = **5 ms**
- Data Link Layer delay = **2 ms**
- Physical Layer delay = **1 ms**
- Propagation delay = **18 ms**
- Transmission delay = **12 ms**

a) Total Processing Delay

$$4 + 3 + 5 + 2 + 1 = 15 \text{ ms}$$

Total Processing Delay = 15 ms

b) Total End-to-End Delay

$$\begin{aligned} \text{Total Delay} &= \text{Processing Delay} + \text{Propagation Delay} + \text{Transmission Delay} \\ &= 15 + 18 + 12 = 45 \text{ ms} \end{aligned}$$

Total End-to-End Delay = 45 ms

c) Contribution of Each OSI Layer

- **Application Layer (4 ms):** Provides network services and prepares user data for transmission.
- **Transport Layer (3 ms):** Segments data, performs flow control, and ensures reliable delivery.
- **Network Layer (5 ms):** Determines the routing path and forwards packets to the destination.
- **Data Link Layer (2 ms):** Frames the data, performs error detection, and controls access to the transmission medium.
- **Physical Layer (1 ms):** Converts data into electrical, optical, or radio signals and transmits them over the physical medium.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Given:

- Useful data = **80 MB = 80,000,000 bytes**
- Frame size = **1600 bytes**
- Protocol overhead per frame = **80 bytes**

a) Useful Data per Frame

$$\text{Useful Data per Frame} = 1600 - 80 = 1520 \text{ bytes}$$

b) Number of Frames Required

$$\text{Number of Frames} = \left\lceil \frac{80,000,000}{1520} \right\rceil = \lceil 52631.58 \rceil = 52,632$$

Number of Frames = 52,632

c) Total Overhead Transmitted

$$\text{Total Overhead} = 52,632 \times 80 = 4,210,560 \text{ bytes}$$

Total Overhead = 4,210,560 bytes ($\approx$ 4.21 MB)

d) Transmission Efficiency

Total transmitted data:

$$80,000,000 + 4,210,560 = 84,210,560 \text{ bytes}$$

$$\text{Efficiency} = \frac{80,000,000}{84,210,560} \times 100 \approx 94.99\%$$

Transmission Efficiency $\approx$ 95%

e) Effect of Protocol Overhead on Network Performance

Protocol overhead is the extra information (headers and trailers) added by different OSI layers to ensure reliable communication.

- Increases the total amount of transmitted data.
- Consumes additional bandwidth.
- Reduces transmission efficiency.
- Enables addressing, routing, error detection, and reliable data delivery.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Given:

- Original medical record size = **240 MB = 240,000,000 bytes**
- Compression = **30%**
- Segment size = **1400 bytes**
- Data Link Layer overhead = **60 bytes/frame**

a) Compressed File Size

Compression reduces the file size by **30%**.

$$\begin{aligned}\text{Compressed Size} &= 240 \times (1 - 0.30) = 240 \times 0.70 = 168 \text{ MB} \\ 168 \text{ MB} &= 168,000,000 \text{ bytes}\end{aligned}$$

Compressed File Size = 168 MB

b) Total Number of Segments

$$\text{Number of Segments} = \frac{168,000,000}{1400} = 120,000$$

Total Number of Segments = 120,000

c) Total Protocol Overhead

Each frame adds **60 bytes** of overhead.

$$\text{Total Overhead} = 120,000 \times 60 = 7,200,000 \text{ bytes}$$

Total Protocol Overhead = 7,200,000 bytes (7.2 MB)

d) Final Amount of Data Transmitted

$$\begin{aligned}\text{Final Data} &= 168,000,000 + 7,200,000 = 175,200,000 \text{ bytes} \\ &= 175.2 \text{ MB}\end{aligned}$$

Final Data Transmitted = 175.2 MB (175,200,000 bytes)

e) How Multiple OSI Layers Work Together

- **Presentation Layer:** Compresses (and can encrypt) the medical record to reduce size and improve security.

- **Transport Layer:** Divides the compressed data into segments and ensures reliable end-to-end delivery through sequencing, acknowledgments, and flow control.
- **Data Link Layer:** Encapsulates each segment into frames, adds headers/trailers, and performs error detection.
- **Physical Layer:** Transmits the framed data as electrical, optical, or wireless signals over the network.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well structured answers with logical flow	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers and coherence

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

